EXPLOITS UNIVERSITY

BSc INFORMATION COMMUNICATION AND TECHNOLOGY

BICT 3202 · OBJECT-ORIENTED ANALYSIS AND DESIGN

WEEK 15

Assignment: Complete OOAD Case Study, Model Integration and CASE Tool Implementation

PROGRAMME	BSc Information Communication and Technology
COURSE CODE / YEAR	BICT 3202 · Year 3
LECTURER	Francis Fweta — ICT Lecturer
DELIVERY	2h Lecture · 1h Tutorial · 1h Practical · 10h Independent Learning

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Assignment Instructions

- Answer ALL questions in Sections A, B and C.
- Read the Week 15 module and your lecture notes before attempting each question.
- Use the university course registration case study where stated.
- Show that your models are consistent: sequence messages must match class operations.
- Draw UML diagrams in correct notation, or with a CASE/UML tool.
- Submit by the date given by your lecturer. Total: 50 marks.

Section A — Short-Answer Questions (20 marks)

Question A1

Define a CASE tool and state three ways it supports software development. [3 marks]

Question A2

Differentiate functional from non-functional requirements, with one example each. [3 marks]

Question A3

Define an actor and explain why an actor need not be a human being. [2 marks]

Question A4

Differentiate a domain class from a technical/application class. [3 marks]

Question A5

State the multiplicity meaning of Student 1 — 0..* Registration. [2 marks]

Question A6

Why should UML diagrams not be developed independently? [3 marks]

Question A7

Explain model traceability using the Register Course requirement. [2 marks]

Question A8

List the six consistency checks of a model audit. [2 marks]

Section B — Application Questions (20 marks)

Question B1 (12 marks)

For university online course registration, a student selects an open course. The system checks prerequisites and active duplicates, atomically reserves capacity and persists the registration in one transaction, then sends confirmation only after commit. On failure the transaction leaves no registration or seat claim.

- (a) Identify the actors and functional requirements. [2 marks]
- (b) Identify candidate domain classes and technical classes. [2 marks]
- (c) Draw the sequence diagram for Register Course, including a rejection path. [3 marks]
- (d) Draw the state machine for Course (Created, Available, Full, Closed). [2 marks]
- (e) Explain how the sequence messages map to class operations. [3 marks]

Question B2 (8 marks)

The activity diagram contains Check Payment, but no payment rule is stated and the class model has no payment-related operation, class or component.

- (a) Identify the modelling problem. [2 marks]
- (b) Explain how to reconcile the models with the approved requirements. [3 marks]
- (c) Describe how a CASE tool helps manage this change. [3 marks]

Section C — Case Study (10 marks)

A college wants an Online Library Management System. Students search, borrow, return and view borrowed books. Librarians add or withdraw physical copies, register members and view overdue loans. A Copy is Available until issued, OnLoan while borrowed, and becomes Available on undamaged return; a damaged return enters Maintenance, then becomes Available when repaired. A copy may be Withdrawn only when not on loan. Each Loan links one member and one copy; a copy can have loan history but at most one active loan.

- (a) Identify actors, use cases and candidate classes. [3 marks]
- (b) Draw a sequence diagram for Borrow Book. [3 marks]
- (c) Draw a state machine for Copy using the stated lifecycle. [2 marks]
- (d) Explain how use-case, class, sequence, activity and state-machine diagrams relate to each other. [2 marks]

Marking Scheme

Section	Content	Marks
A	Short-answer questions (8 questions)	20
B	Application questions (12 + 8)	20
C	Library management system	10
	TOTAL	50

Submission & Academic Integrity

- Submit a typed document (or neatly handwritten, as directed by your lecturer).
- This is individual work. Plagiarism or copying attracts a penalty under university regulations.

- Use correct UML notation, including multiplicities, messages, events and guards.
- Cite sources using the format in the Week 15 module references.